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Comparative Analysis of Selected Characteristics of Java and Rust Based on Programs Implementing Equivalent Functionalities

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Introduction

Choosing the right programming language for implementing a solution to a specific problem is highly important. First, we need to balance application performance and the cost of implementation. Some tasks require fast execution time and limited resources, while others may be implemented with fewer restrictions. One of the most important aspects that should be considered is the cost of development. In particular, a project often has financial and time constraints that must be fulfilled. On the other hand, from the perspective of software engineering, the codebase should be as easily maintainable as possible.

Such limitations have a great impact on programming language selection. Therefore, in this study, we perform a comprehensive analysis and comparison of two popular programming languages. Java represents a high level of language abstraction, while Rust, in contrast, constitutes a low-level programming language.

Research Scenarios

1. *krótki opis*
2. *którki opis*

- *wniosek 1*
- *wniosek 2*

Experimental Results

Figure

Figure 1. Execution Time.
a) CPU, b) IoT Streaming, c) I/O operations

Figure

Figure 3.Garbage Collector Pauses Over Time
a) IoT Streaming, b) I/O operations

Figure

Figure 2. Memory Usage Over Time.
a) CPU, b) IoT Streaming, c) I/O operations

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